

Module 4

Signals, Emissions & Practical Circuits

General License Course · G7 + G8 · 6 Exam Questions



Hour 1

Practical Circuits: Power, Amplifiers, Receivers



G7A — Power Supplies

Transformer: steps AC voltage up or down; provides isolation from the power line (safety)

Rectifier: converts AC to pulsating DC; common types: half-wave (wastes half), full-wave (center tap), full-wave bridge (most common)

Filter capacitor: smooths pulsating DC to ripple DC; larger capacitor = less ripple

Ripple frequency: for full-wave rectifier on 60 Hz AC → ripple = 120 Hz (twice the input frequency)

Voltage regulator: maintains constant output despite load or line changes; IC regulators (7805 = 5V, 7812 = 12V, etc.)

Switching power supply: chops DC at high frequency for efficiency; compact and efficient; can cause RFI

Linear supply: wasteful (drops excess voltage as heat) but very clean RF-wise

Battery types: lead-acid (flooded, AGM, gel), lithium iron phosphate (LiFePO₄); charging requirements differ



G7B — Amplifier Classes

Linear Classes (for SSB)

Class A: always conducting; output follows input; low distortion; ~25-30% efficient

Class AB: conducts slightly more than half cycle; good linearity; 50-70% efficient; most SSB finals use this

Both A and AB required for SSB — you need linear amplification to preserve the sideband's information

Non-Linear Classes (CW/FM)

Class B: conducts exactly half cycle; more efficient than A (~75%)

Class C: conducts less than half cycle; ~80% efficient; highly non-linear; only for CW or FM where distortion doesn't matter

Class D/E: switching amplifiers; very high efficiency; emerging in ham radio

Class C on SSB = terrible distortion = splatter = causing interference to adjacent channels



G7B — Negative Feedback in Amplifiers

Negative feedback: take a fraction of the output, invert it, add it back to the input

Effect: the amplifier "corrects" itself → less distortion → more linear response

Trade-off: gain is reduced by the feedback amount (if you feed back 10%, gain drops $\sim 10\times$)

Benefits: more stable gain over temperature and frequency; lower output impedance; wider bandwidth

Op-amps run with massive negative feedback (open-loop gain = 100,000, closed-loop = whatever you set with resistors)

Oscillators: use POSITIVE feedback → signal grows until limited; the feedback must be in-phase

The difference: negative feedback stabilizes; positive feedback creates oscillation

Audio amplifiers in transceivers always use negative feedback for low distortion



G7C — The Superheterodyne Receiver

Superheterodyne (superhet): the architecture used in virtually all HF receivers since the 1920s

Signal path: Antenna → RF stage → Mixer → IF amplifier → Detector → Audio amplifier → Speaker

Mixer: combines the incoming RF with a local oscillator (LO) signal; produces sum and difference frequencies

IF (Intermediate Frequency): a fixed frequency (455 kHz common for AM; 10.7 MHz for FM; 8.215 MHz for SSB); filtering happens here

Why convert to IF? A fixed-frequency filter is much easier to make sharp and stable than a tunable one

BFO (Beat Frequency Oscillator): injects a carrier to heterodyne against the IF; required for SSB and CW demodulation

AGC (Automatic Gain Control): maintains constant audio level as signal strength varies; samples IF level

SDR: Software Defined Radio — ADC digitizes the signal early in the chain; everything else done in software



Hour 2

Signals and Emissions



G8A — Modulation Modes Compared

AM (Double-sideband AM): carrier + two sidebands; carrier wastes 2/3 of transmitter power

SSB (Single Sideband, suppressed carrier): one sideband, no carrier; most efficient; ~3 kHz bandwidth

FM (Frequency Modulation): frequency varies with audio; constant amplitude (no amplitude fading); ~15 kHz for VHF

Emission designators: J3E = SSB phone (the J means suppressed carrier, 3 = phone, E = telephony)

A3E = AM double sideband; F3E = FM phone; G3E = PM phone; F1B = RTTY; J2B = PSK

Bandwidth comparison: CW (~200 Hz) < PSK31 (~80 Hz, narrower!) < SSB (~3 kHz) < AM (~6 kHz) < FM (~15 kHz)

Why SSB for HF? Efficient power use, narrow bandwidth, punches through noise better than AM

Why FM for VHF/UHF? Noise immunity (capture effect), simple receivers, FM-only repeaters



G8B — FM Deviation and Modulation

FM deviation: how far the carrier swings above and below center frequency in response to audio

Amateur VHF FM: ± 5 kHz maximum deviation (so channel width = $2 \times 5 = 10$ kHz around center)

Modulation index (FM): ratio of frequency deviation to modulating frequency; index = $\Delta f / f_m$

Deviation ratio: ratio of MAXIMUM deviation to MAXIMUM modulating frequency

Overdeviation: audio too loud $\rightarrow$ exceeds ± 5 kHz $\rightarrow$ adjacent channel interference; "wideband" sounds of over-deviated stations

Capture effect: FM receiver "captures" the stronger of two co-channel signals; suppresses the weaker; why FM is good in mobile use

Intermodulation products: $f_{IM} = n \times f_1 \pm m \times f_2$; third-order ($2f_1 - f_2$) fall close in — receiver problem

Carson's rule: approximate FM bandwidth = $2 \times (\text{max deviation} + \text{max modulating freq})$



G8C — Digital Modes

RTTY (Radio Teletype): two frequencies (mark and space), 170 Hz shift standard, 45.45 baud; uses LSB with AFSK

PSK31: phase-shift keying; 31.25 baud; ~80 Hz bandwidth; keyboard-to-keyboard; uses LSB with AFSK

PSK31 note: dial frequency $\neq$ RF frequency; add audio tone (~1500 Hz) to dial to get actual RF

FT8: 15-second transmit/receive cycles; computer clock must be within ± 1 second of UTC; 4-tone GFSK; ~50 Hz wide

FT8 uses: WSJT-X software; exchanges grid squares and signal reports; fully automated QSO

MFSK: multi-frequency shift keying; multiple tones for higher throughput; more robust than 2-tone FSK

Winlink: email over radio; HF gateways; uses PACTOR or Ardop; addressed as regular email

APRS: 1200 baud on VHF (144.39 MHz); position, messaging, weather; uses AX.25 protocol



Checkpoint Quiz

What is the ripple frequency output of a full-wave rectifier connected to a 60 Hz AC source?

A. 120 Hz

B. 60 Hz

C. 240 Hz



Checkpoint Quiz

Which amplifier class is most suitable for use in the final stage of a single-sideband transmitter?

A. Class AB

B. Class A

C. Class C



Checkpoint Quiz

What term describes the ability of an FM receiver to suppress the weaker of two co-channel signals?

A. Capture effect

B. Quieting effect

C. Desensitization



Checkpoint Quiz

What is the emission designator for a single-sideband phone signal with a suppressed carrier?

A. J3E

B. A3E

C. F3E



Checkpoint Quiz

What function does a Beat Frequency Oscillator (BFO) serve in a receiver?

A. It generates a carrier to heterodyne with the IF signal for SSB and CW demodulation

B. It stabilizes the local oscillator frequency

C. It provides automatic gain control for the IF stage



HamStudy Assignment

Drill these sections before next module:

G7A — Power Supplies

G7B — Amplifiers & Oscillators

G7C — Receivers & Transmitters

G8A — Modulation Modes

G8B — FM & Modulation

G8C — Digital Modes

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